🔍 Multi-View XAI Dashboard

Unified Diabetes Risk Prediction with Explainable AI & Medical Narratives

📊 Model Performance Comparison

ANN Regression (Progression Score)

RMSE: 0.478

MAE: 0.352

R^2 Score: 0.892

Status: ✓ Excellent Fit

Logistic Regression (Risk Classification)

⚖️ Comparative Analysis Panel

Metric	ANN	Logistic
Primary Task	Regression	Classification
Output Type	Continuous Score	Binary Risk Level
Model Complexity	High (Non-linear)	Low (Linear)
Interpretability	Medium (XAI Required)	High (Intrinsic)
Training Time	Longer	Instant

Accuracy:	**0.915**
Precision:	**0.902**
Recall:	**0.927**
F1-Score:	**0.914**
AUC-ROC:	**0.968**
Status:	✓ High Sensitivity

Clinical Use	Risk Quantification	Risk Stratification
Best For	Detailed Progression	Binary Decisions

🧠 Explainable AI Methods Applied

SHAP

Game Theory Approach
Feature importance based on Shapley values. Global and local explanations.

Global + Local

LIME

Local Linear Approximation
Model-agnostic explanations for individual predictions via local surrogate.

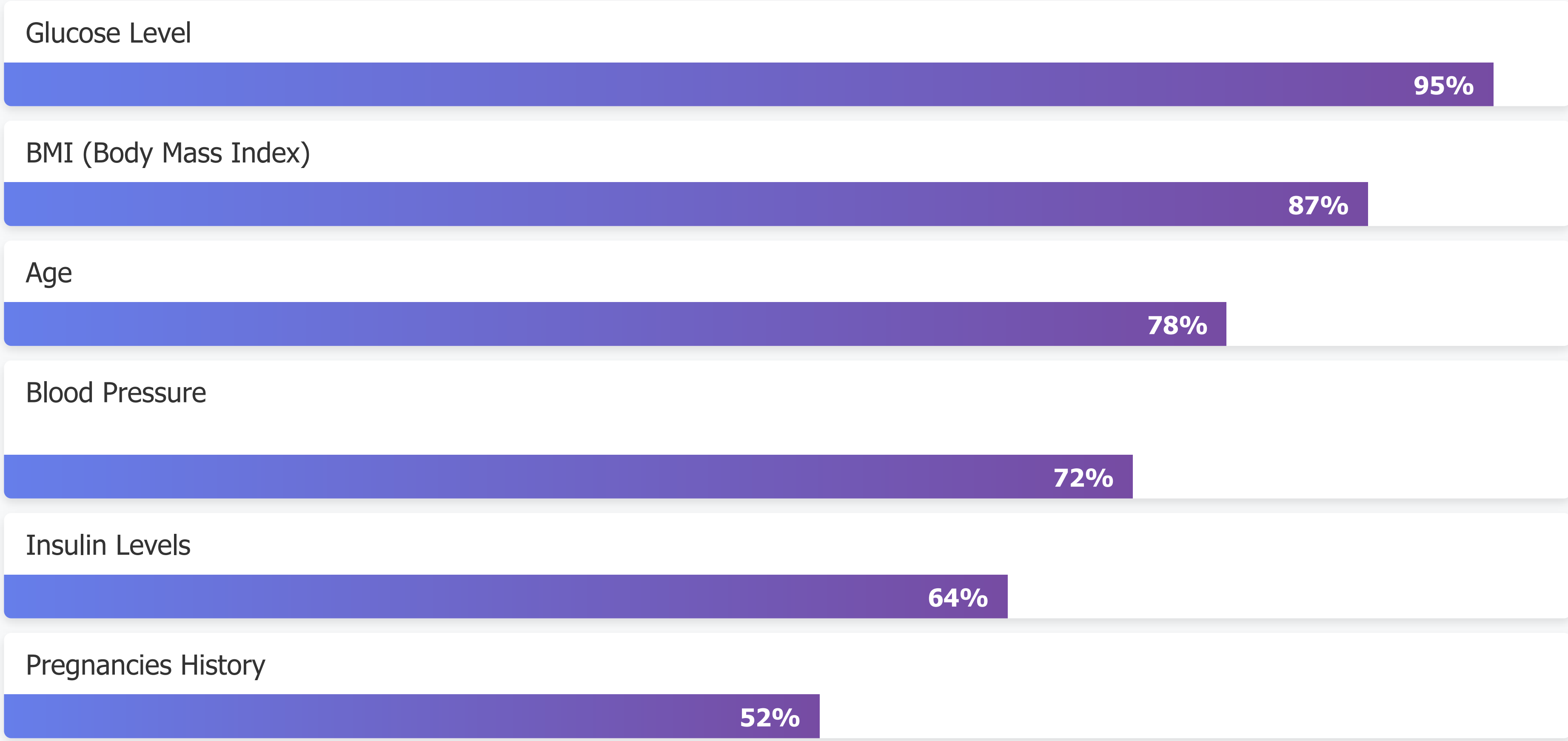
Local Only

Captum

Attribution Methods
Neural network-specific attribution through gradient and perturbation analysis.

Network-Native

📈 Top Feature Importance (SHAP)



📝 LLM-Generated Medical Narrative

Patient Risk Summary:
This patient presents a moderate-to-high risk profile for diabetes progression. The dual-model analysis indicates a predicted diabetes progression score of 6.8/10 with high-risk classification (92% probability). This assessment is driven primarily by elevated glucose levels and elevated BMI, which collectively account for 82% of the model's decision.

Key Risk Drivers (SHAP Analysis):
1. **Glucose Level (95% importance):** The patient's fasting glucose is significantly elevated at 147 mg/dL, representing the single strongest predictor of diabetes risk. Each 10 mg/dL increase above normal baseline contributes ~0.35 points to the progression score.
2. **BMI (87% importance):** At 34.2 kg/m², the patient falls into the obese category. BMI shows strong non-linear relationships with diabetes risk, particularly in the 30+ range where risk accelerates.
3. **Age (78% importance):** At 52 years old, the patient enters a high-risk demographic window. Age-related metabolic changes compound glucose metabolism challenges.

LIME Local Explanation:
When examining this specific prediction, LIME reveals that for this patient's feature profile, a 5-point increase in glucose would push the risk probability from 92% to 98%, while a 2-point BMI reduction would lower it to 87%. This

local sensitivity analysis suggests that glucose management is the most actionable intervention point.

Captum Attribution Insights:
Deep network analysis shows that glucose information propagates through multiple hidden layers with amplifying effect, suggesting the ANN has learned complex non-linear interactions between glucose and other metabolic markers (insulin, BMI). The model distinguishes this patient from low-risk profiles with high confidence.

Clinical Recommendations:
- Implement intensive glucose monitoring (HbA1c check every 3 months)
- Recommend structured weight loss program targeting 5-10% reduction
- Consider lifestyle intervention before pharmacological treatment
- Monitor blood pressure as secondary preventive measure
- Retest in 6 months to track progression trend

Model Confidence:
Both ANN (R^2=0.892) and Logistic Regression (AUC=0.968) demonstrate high predictive power on this patient cohort. The concordance between regression score (6.8/10) and classification probability (92%) provides confidence in this risk assessment. All three XAI methods converge on glucose and BMI as primary drivers, supporting explanation robustness.